

FINAL PROJECT • DYNAMICAL MODELS FOR SOCIAL BEHAVIOR

Bebop as a Markov Chain

A discrete-time finite-state random walk on a graph of 428 states, estimated from Charlie Parker's improvisational lines.

Research question. *Can the syntactic rules of bebop improvisation (voice leading, harmonic gravity, the on-beat / off-beat asymmetry) be recovered as numerical properties of a Markov-chain operator estimated from Parker's solos?*

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Course: *Dynamical Models for Social Behavior* • Profs. Gonnella & Suma

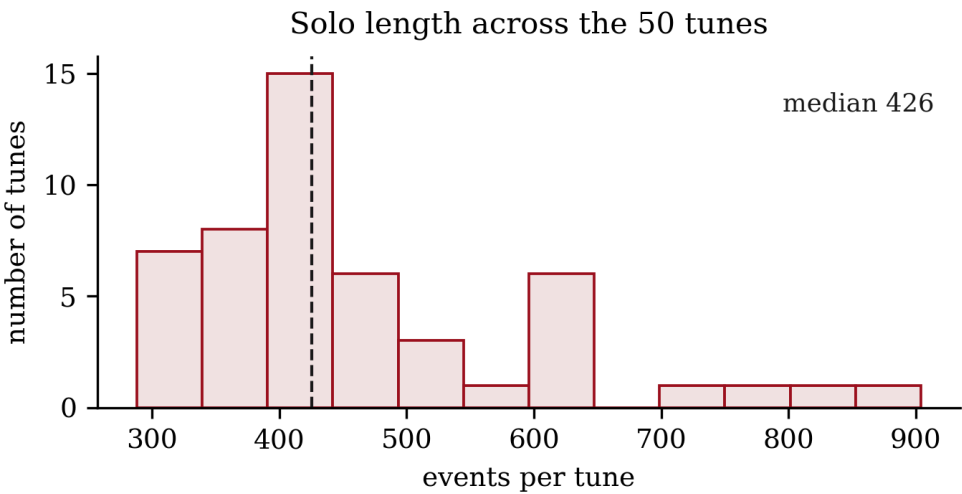
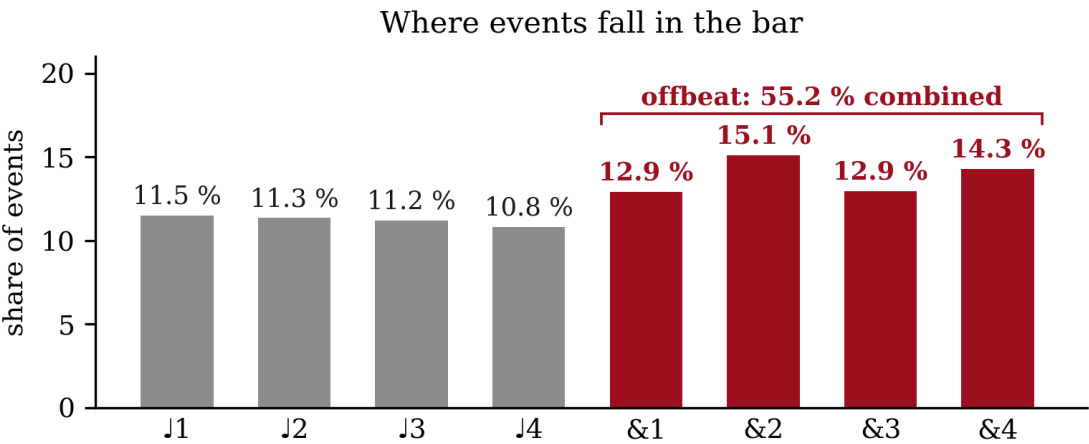


50 Charlie Parker improvs 23,143 events (21,590 notes + 1,553 rests)

- Source: MusicXML transcriptions of solos from the Parker Omnibook.
- Each XML file = one tune composed of head (exposed melody) + one or more improvised choruses, with harmony annotations of active chord at every event.
- Extraction: walk each part, pair each melodic note with the most recent chord, collapse tied notes to a single event.

Chord quality	notes	%
dom7	12,019	55.7
min7	5,397	25.0
maj7	3,862	17.9
m7 b 5	157	0.7
dim7	155	0.7

Corpus statistics



State = (scale degree, chord quality, beat position)

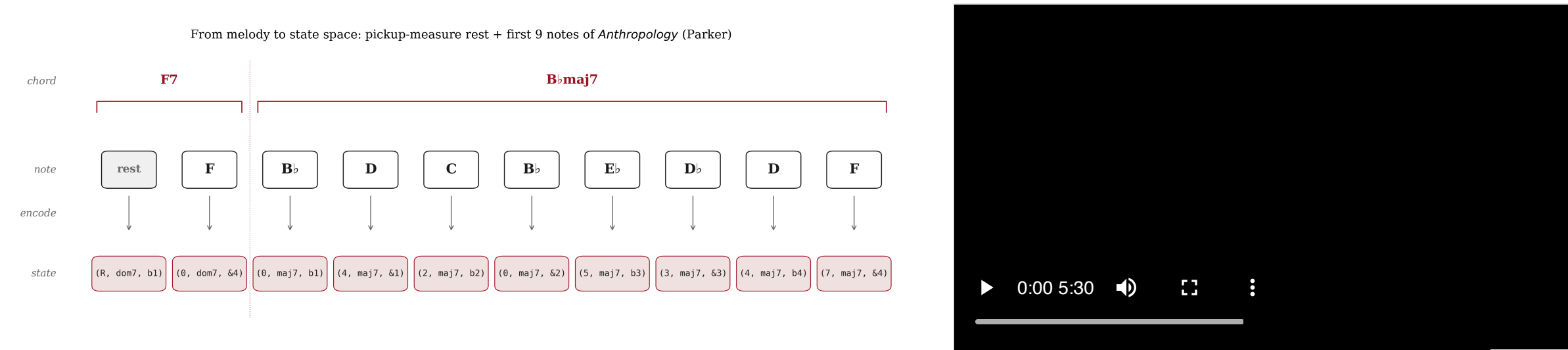
Three discrete coordinates per event (note or rest):

- **Scale degree** $d \in \{0, \dots, 11\} \cup \{R\}$: for a note, semitones above the active chord root $d = (pc_{\text{note}} - pc_{\text{chord-root}}) \bmod 12$; for a rest, the sentinel R.
- **Chord quality** $q \in Q = \{\text{maj7}, \text{min7}, \text{dom7}, \text{m7}\flat 5, \text{dim7}\}$: the active chord at event onset (rests inherit it).
- **Beat position** $b \in \{b1, b2, b3, b4, \&1, \&2, \&3, \&4\}$: all 50 tunes are 4/4, so an event either lands on one of the four beats or falls offbeat, labelled by the beat it follows.

Why this state space

- **Scale degree** abstracts over transpositions: the same line over F7 and over B $\flat$ 7 collapses to one state.
- **Chord quality** respects the functional context: a " $\flat$ 7" sounds like the dominant 7th, the half-diminished 7th, or the dim7 depending on quality.
- **Beat position** separates structural notes (downbeats) from passing material (see bebop's passing notes).

Anthropology, bar by bar



Result

$|S| \leq 13 \times 5 \times 8 = 520$; observed: **428**. The encoding collapses **23,143 events** onto those 428 states: about **54 visits per state**, and every tune revisits most of its states several times over. That recurrence is what makes the transition probabilities estimable at all.

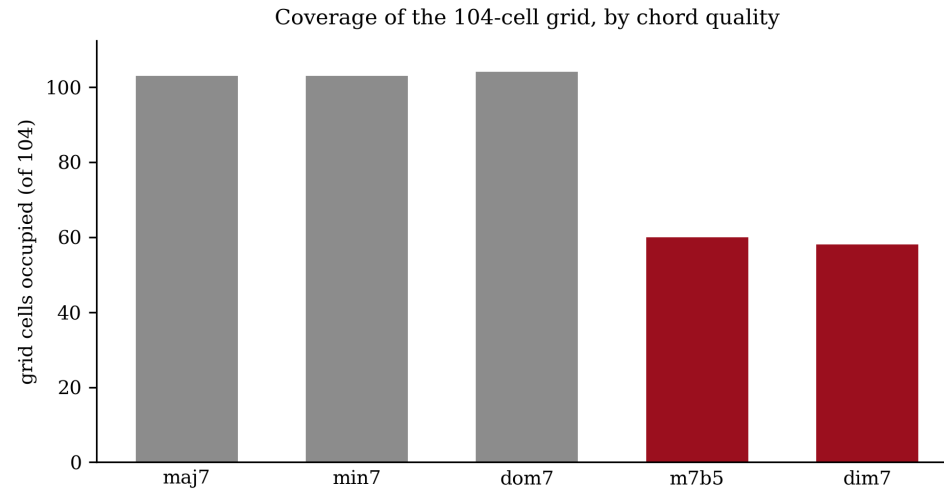
Important: we are not considering durations, octaves (heights), dynamics (f, p, pp) etc.

What Parker never plays

428 of 520 states occur. The **92** that **never do** split into two different causes.

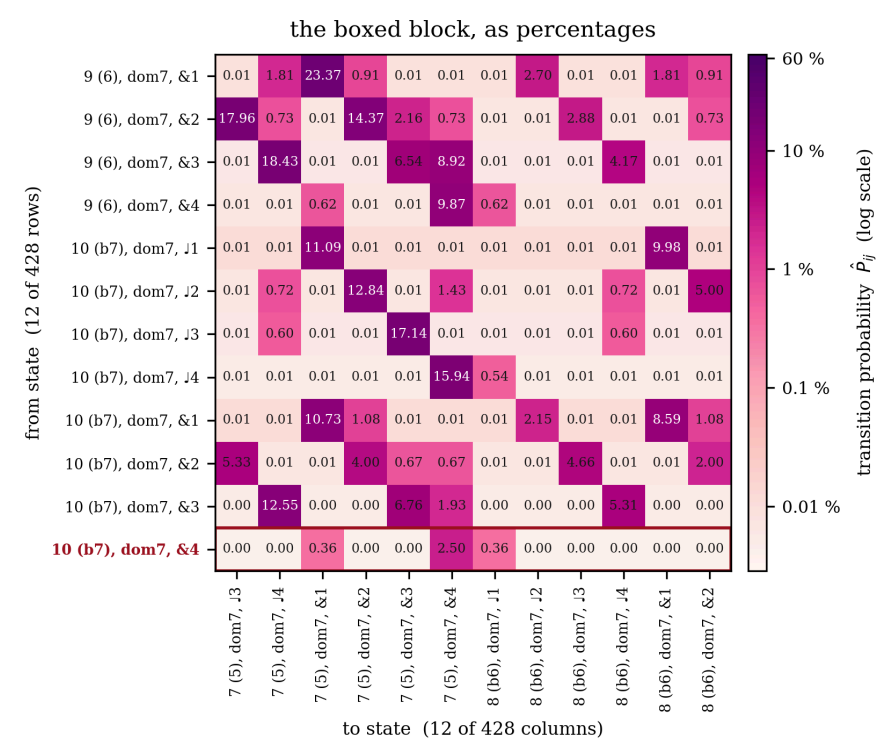
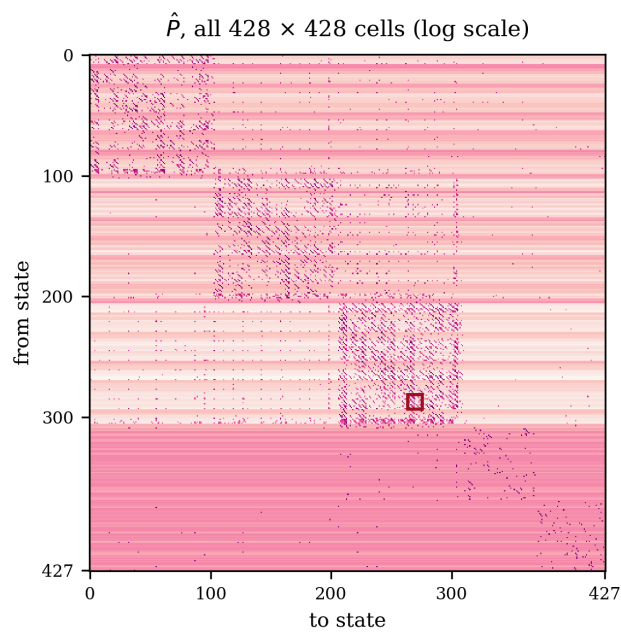
Scarcity, 90 cells. Each quality has its own $13 \times 8 = 104$ -cell grid. **dom7** fills all 104, but **m7^b5** and **dim7** carry only ~ 165 events each and cannot cover theirs.

Some of the other states are simply never used because they contain a note that **clashes with the chord**, such as a major third over a minor chord.



$\hat{P}$ is a table: 428 rows, one question each

Every row asks one question: **Parker is on this state, where does he go next?** Walk the 23,143 events in order; each time state i is followed by state j , add 1 to cell (i, j) , then divide each row by its own total. The hat marks an **estimate**: $\hat{P}$ is what 50 solos say about the unobservable true P . Probabilities live **within a row**: each row sums to 1 across **all 428 columns**. Much of the emptiness is **metre**: from **&4** the walk goes to **b1** 69 % of the time or stays in **&4** 21 %, and reaches **b3** just 0.4 % of the time. Most cells are empty because the beat cannot get there. Reading the colour: **dark dots** are moves Parker actually plays; **near-white** means the model is confident a move does not happen; **pink** means it has almost no data for that row and is spreading the probability evenly over all 428 states.



The Markov property: is the current state enough?

$$\Pr(X_{t+1} \mid X_0, \dots, X_t = s_i) = \Pr(X_{t+1} \mid X_t = s_i)$$

The next state depends only on where the line is, not on how it got there. Everything before the current event is discarded.

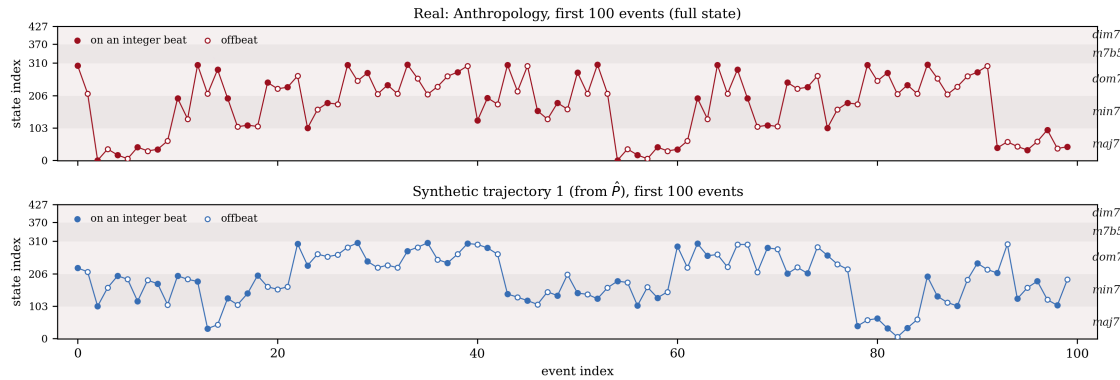
This is the licence for the previous slide. If history mattered, Parker could not be summarised as a table of "from → to".

Take 7 (5), dom7, &4, which occurs 346 times with a known predecessor, and split its continuations by what came *before*:

arrived from	most likely continuation
9 (6), dom7, b4 n = 63	0 (1), dom7, b1 22 %
4 (3), dom7, b4 n = 45	10 (b7), dom7, &4 40 %

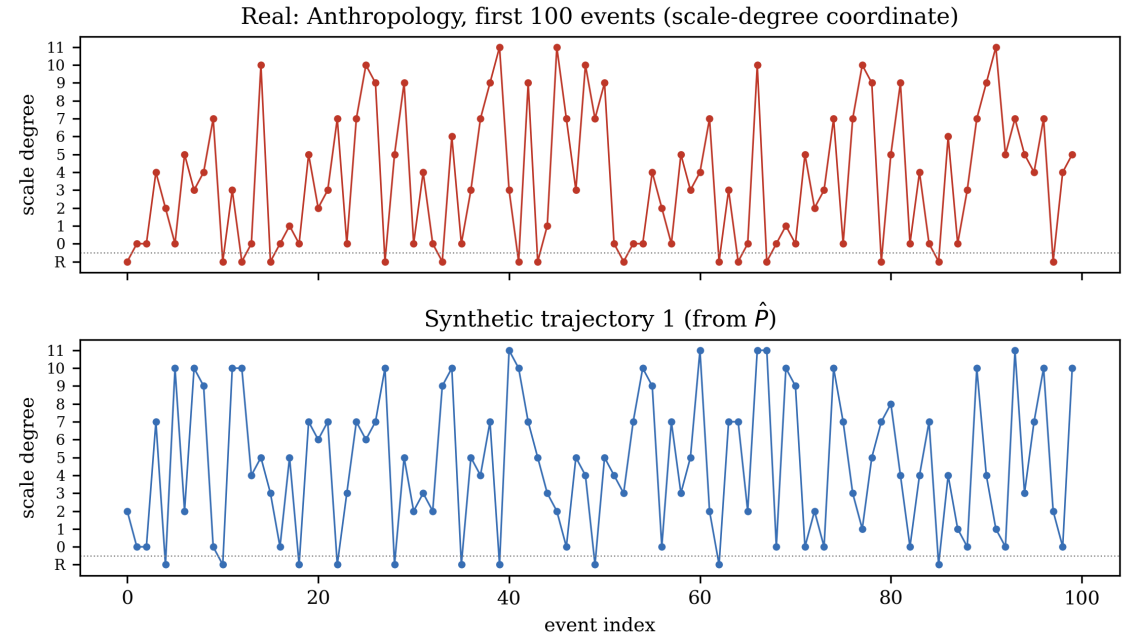
Same current state, different futures. Approached from the 3rd (semitone 4), the line keeps descending to the ♭ 7 (semitone 10) four times out of ten; approached from the 6th, that continuation barely appears. **The property is false in the strict sense**, and we know the direction of the error: descending lines carry momentum a first-order chain cannot see.

A walk sampled from $\hat{P}$, beside Parker's own



The walk itself. Every step is one complete state, plotted by its index. States are ordered by chord quality. Height *within* a band is scale degree; a jump *between* bands is a chord change. Filled marker = on an integer beat, hollow = offbeat.

Over the same 100 events Parker visits **52 distinct states** and the sampled walk visits **81**. The real line keeps returning to a smaller working vocabulary.



The same walk, projected. Two of the three coordinates are discarded and only the scale degree is kept (0 = root ... 11 = major 7th), with **R** for a rest below the dotted rule. This reads as melodic contour, which the panel on the left does not, but it is a shadow: two points at the same height can be completely different states, differing in chord quality or beat.

What the sampled walk sounds like on paper

The image displays two staves of music in 4/4 time, comparing Parker's original performance of 'Anthropology' with a synthetic trajectory. The top staff, labeled 'Parker', shows the original melody with seven measures of music. Above the staff are the following chord changes: F7, Bbmaj7, Cm7, F7, Dm7, G7, Cm7, and F7. The bottom staff, labeled 'Synthetic trajectory 1', shows the same melody but with randomized rest durations. The notes are identical to the original, but the timing of the rests is altered, creating a different rhythmic feel. The synthetic version starts on the downbeat of bar 2, while the original starts on the downbeat of bar 1.

Top: the pickup and first four bars of *Anthropology* as Parker played them. Bottom: the opening events of the **same sampled walk** shown on the previous slide, entering on the downbeat of bar 2 over the identical changes. Rest durations are randomized: the model does not carry them.

Where the walk spends its time

Let the walk run a long time and count how often it is in each state. That settles to a fixed share per state, written π : one number per state, adding to 1.

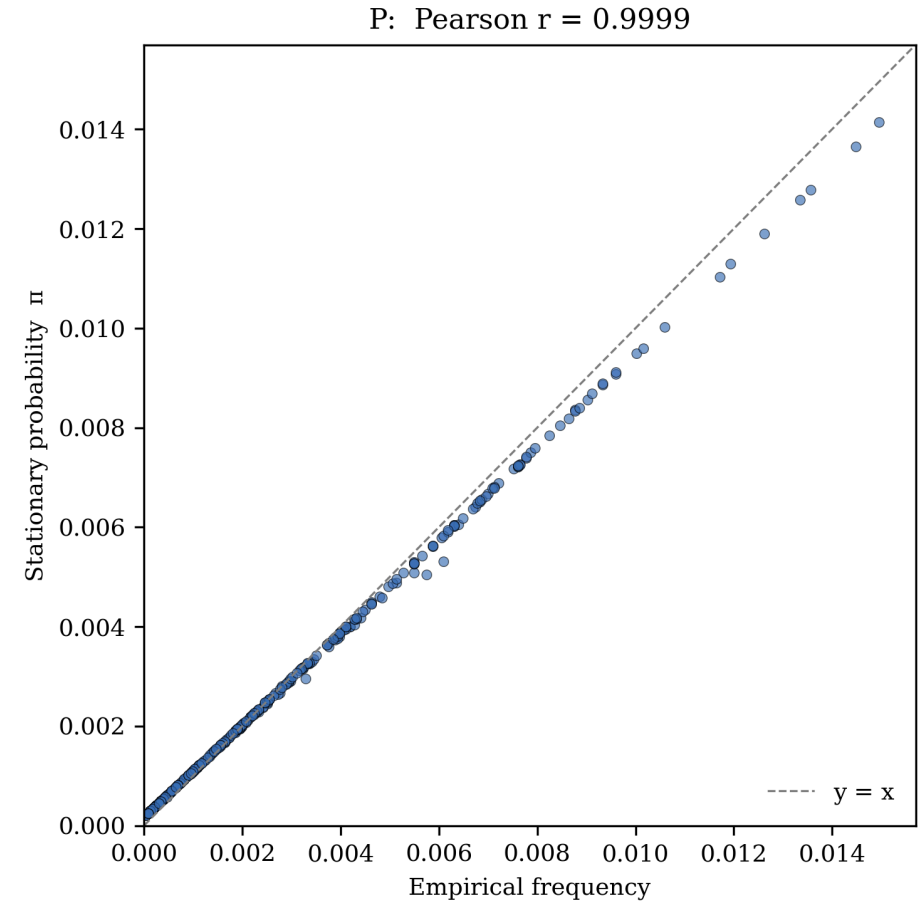
$$\pi \hat{P} = \pi$$

π is the distribution that **one more step leaves unchanged**.

Why that matters. π comes out of the *transitions*, never from how common each state is, yet it reproduces the corpus's own frequencies. A check on the whole estimation.

The three busiest states

- | | |
|--------------------|----------------|
| 1. 7 (5), dom7, &4 | $\pi = 0.0141$ |
| 2. 0 (1), dom7, &4 | $\pi = 0.0137$ |
| 3. 7 (5), dom7, &3 | $\pi = 0.0128$ |

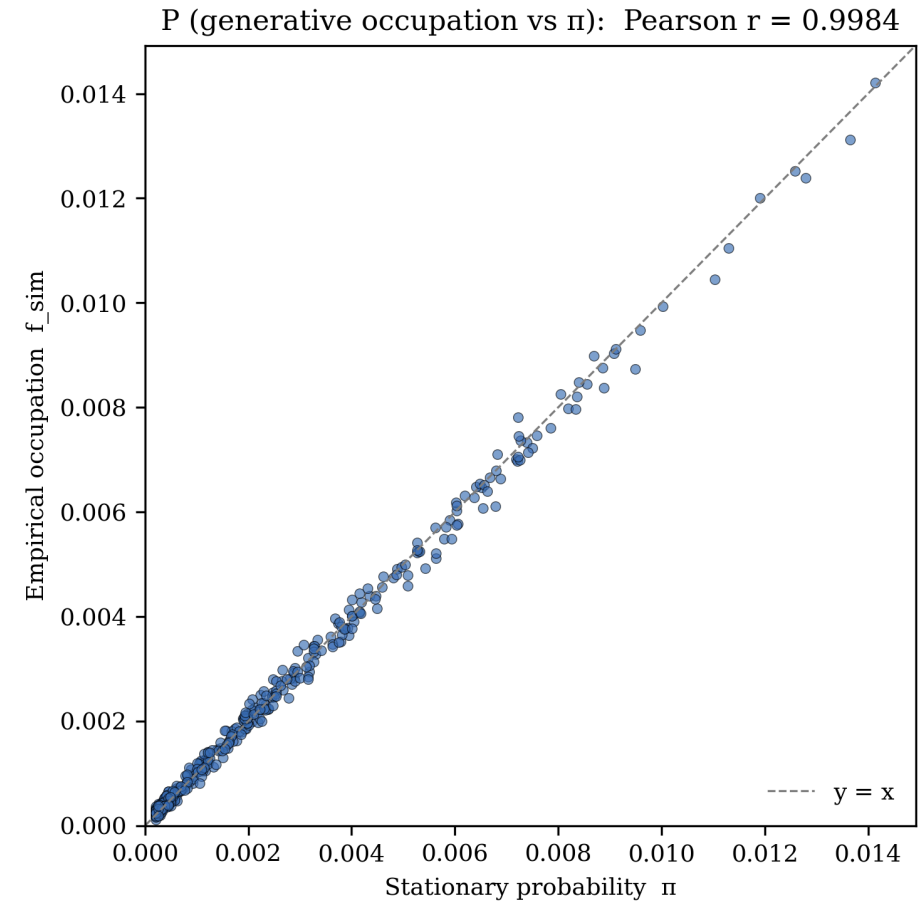


One dot per state: π against how often that state actually occurs. The diagonal is perfect agreement.

Letting agents loose on the graph

Drop **1,000 agents** on random starting states. Each takes **100 steps**, choosing where to go by the odds in its current row. That is **100,000** events of newly generated Parker.

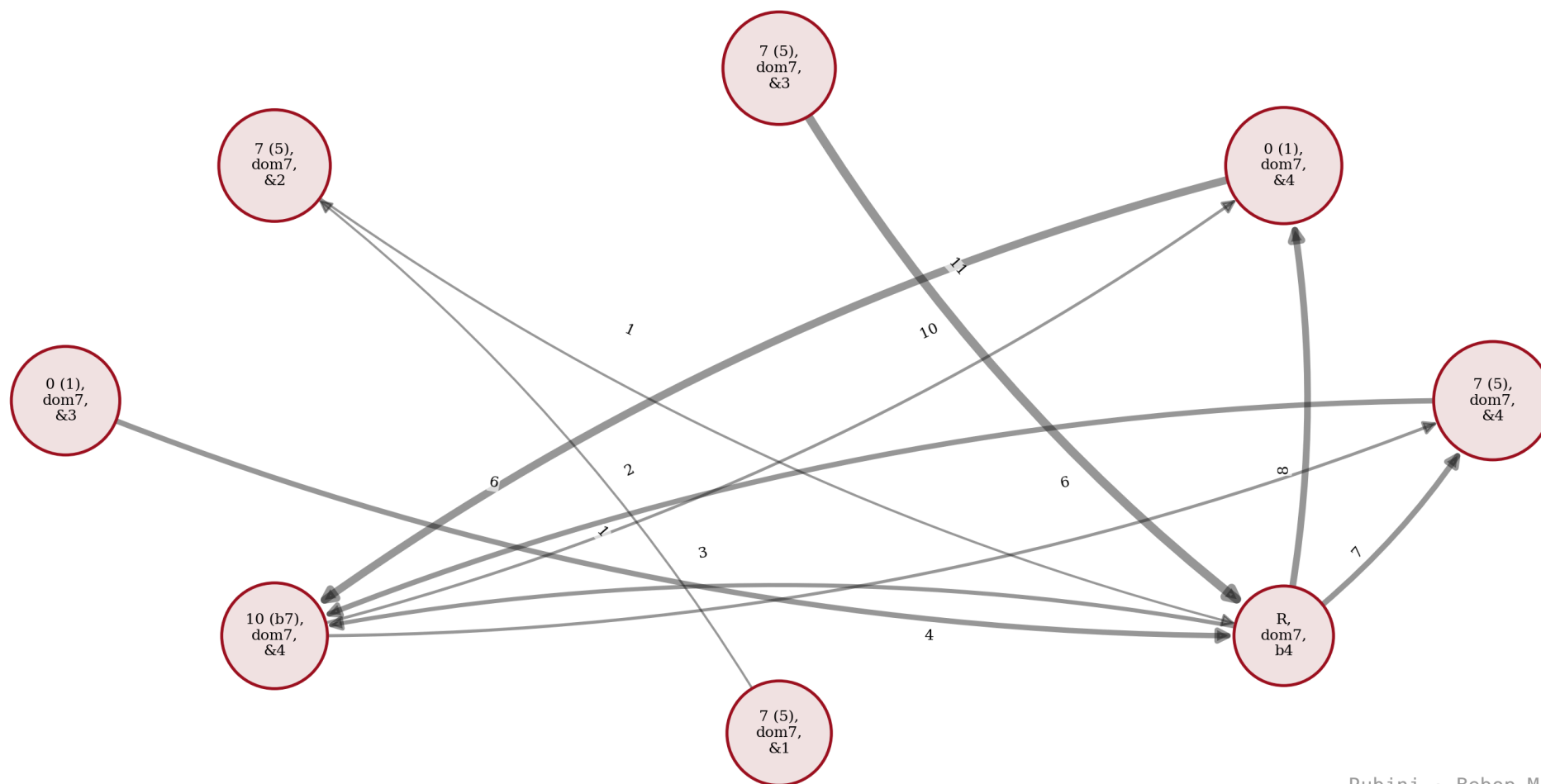
Then count where they spent their time, and compare with π .



Each dot is one of the 428 observed states. Diagonal = perfect match.

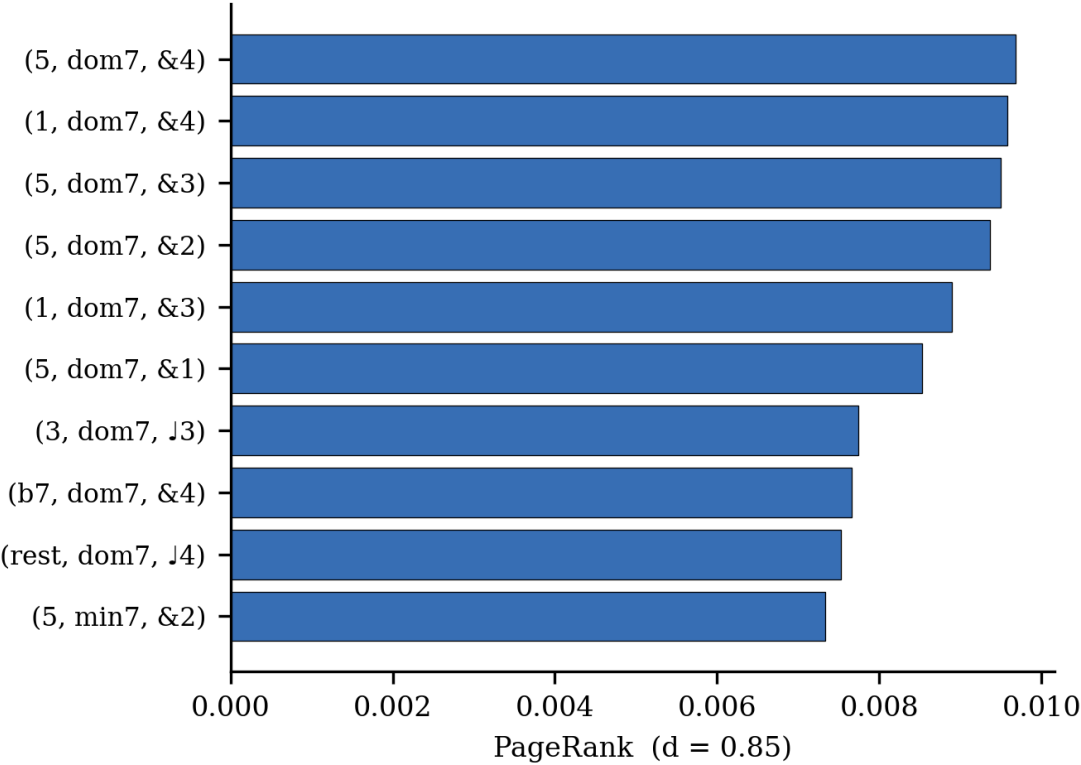
Eight nodes, so the walk is something you can look at

$\hat{P}$ is a 428-node directed weighted graph: nodes are states, edge $i \rightarrow j$ carries weight $\hat{P}_{ij}$. Here is the subgraph on the **eight states with the largest π** , which between them account for **9.7 %** of the walk's time. Every one of the eight is a **dom7** state, and seven of the eight sit offbeat.

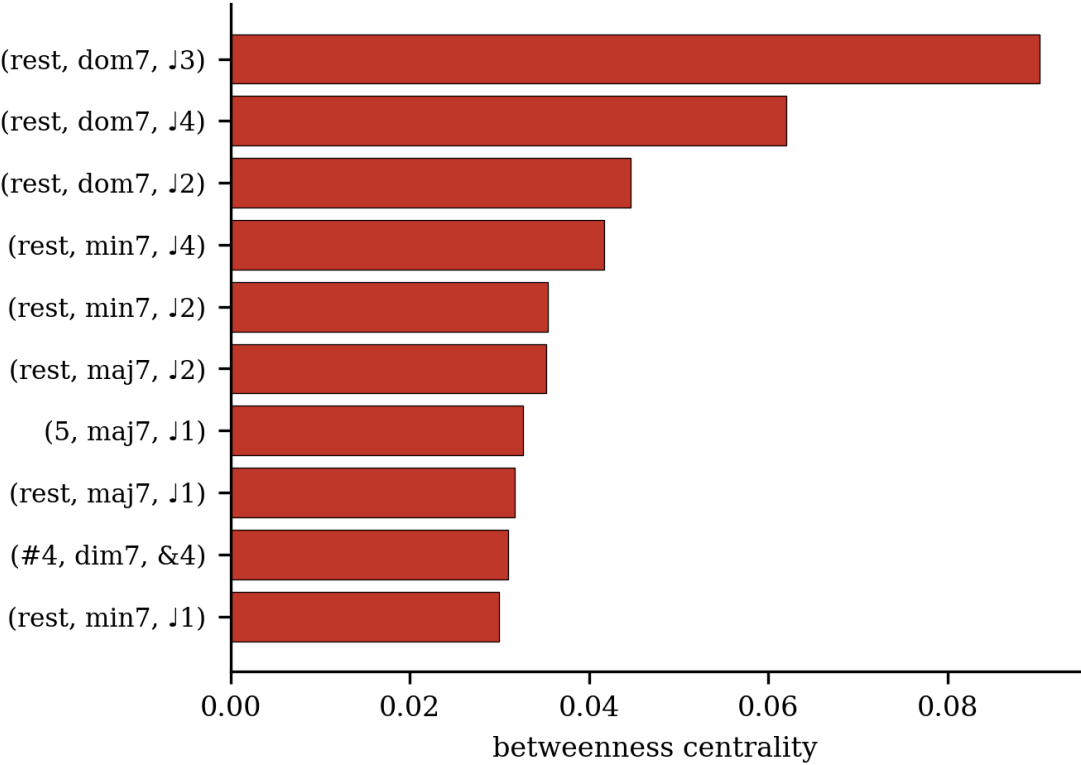


Random surfer with damping

PageRank runs a walk with one change: **85 %** of the time follow an edge, otherwise **jump to a state chosen at random**. The teleport keeps the walk from being trapped anywhere, so the ranking is unique whatever the graph looks like. Alongside it, **betweenness** asks a different question of the same graph: not where the walk spends time, but which states sit on the likely routes between other states. 428 nodes, 5,421 directed edges.



Where the walk spends its time. Nine of the ten are dominant sevenths, eight offbeat.



Which states the routes run through. Eight of the ten are rests, nine on the beat.

Offbeat notes are where the line **lives**; rests on the beat are where it **passes through**.

Musical "gravity" as how long resolution takes

Mark a set of states as **resolved**: a chord tone landing on one of the four beats.
That is **73** of the 428.

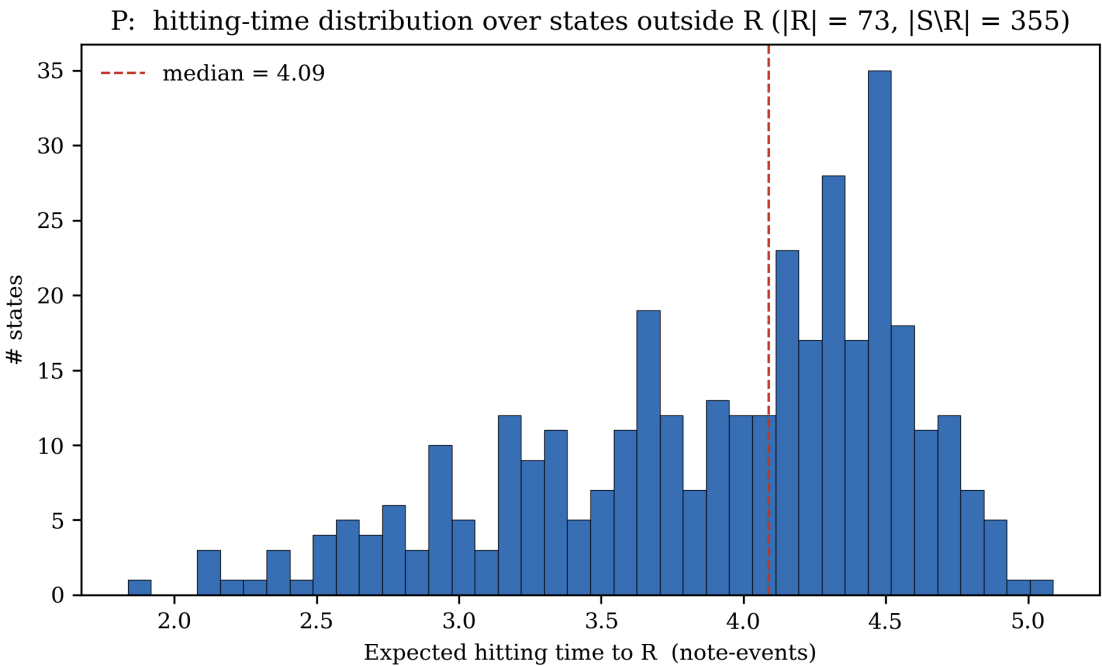
Now ask, from every *other* state: **starting here, how many events pass before the line first lands on one of those?**

typical state	4.09 events to resolve
fastest state	1.84 events
slowest state	5.09 events

What resolves fastest

11 (7), min7, &1	1.84
1 (b2), dom7, &3	2.10
1 (b2), maj7, &4	2.13

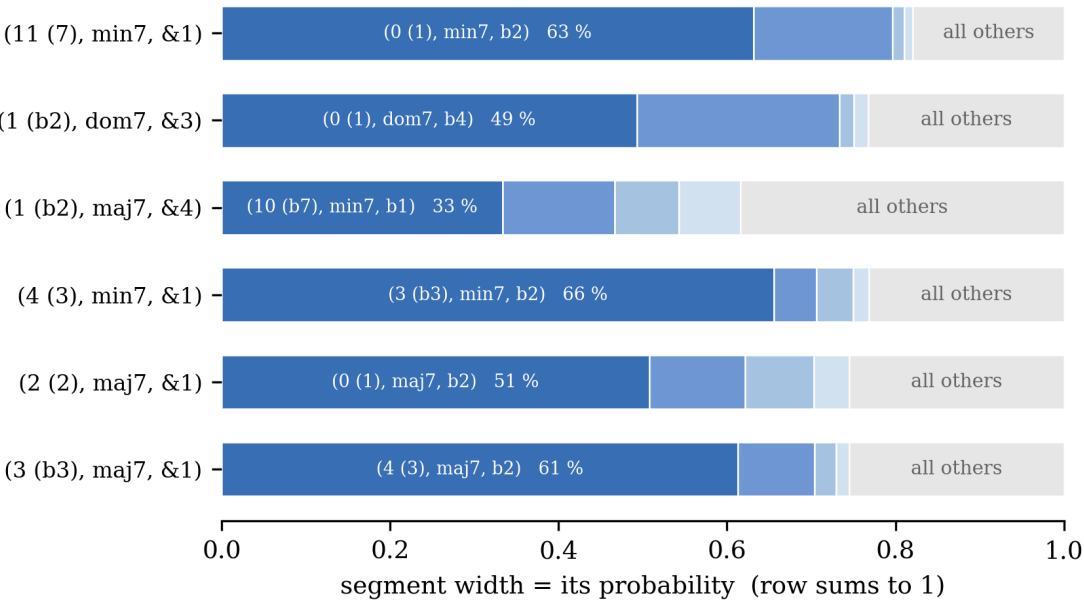
These are the notes theory calls **approach tones**: a semitone from a chord tone, and unstable. The chain finds them without being told which notes clash. **All eight fastest sit offbeat**, so the pull towards resolution points at the downbeat.



Where does the line resolve to?

Take an unresolved note and follow the line forward until it first lands on a chord tone on a beat. Which one did it land on?

- 1. Split the 428 states: R, the 73 resolved ones, and T, the other 355.
- 2. Cut two blocks out of $\hat{P}$: Q, the moves $T \rightarrow T$, and $P_{T \rightarrow R}$, the moves $T \rightarrow R$ in one step.
- 3. Solve $(I - Q) B = P_{T \rightarrow R}$, where I is the identity matrix, the matrix version of the number 1.
- 4. Row i of B is a distribution over R: starting unresolved at i, the chance each resolved state is the **first** one reached. Rows sum to 1.



Four of the six resolve by a semitone onto the nearest chord tone, between 49 % and 66 % of the time. A fifth, the 2nd over a major chord, falls a whole tone to the root. Nothing told the model what a semitone is.

Which moves actually carry the music

The question

A row of $\hat{P}$ can be near-certain about a state Parker hardly ever plays. So **which transitions actually carry the music?** That needs both how often the line is at i and where it goes.

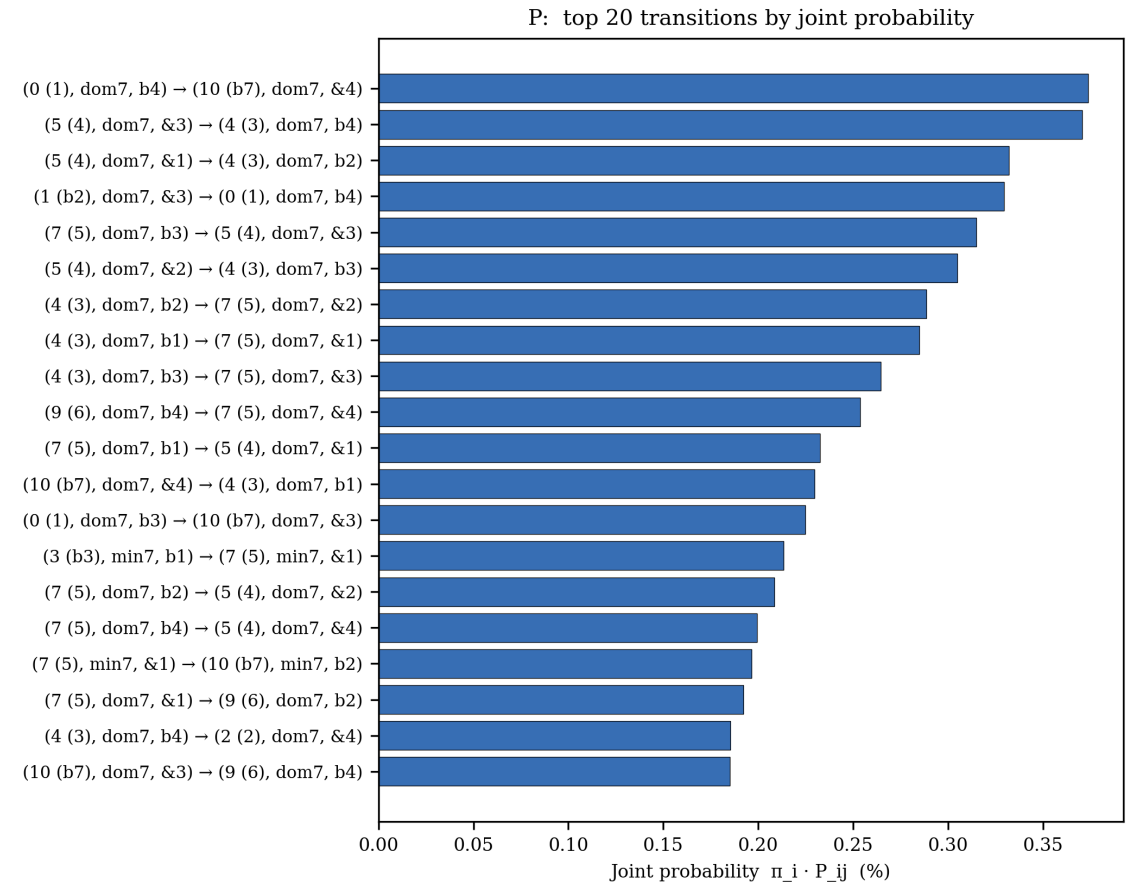
What we compute

$$\Pr(X_t = i, X_{t+1} = j) = \pi_i \cdot \hat{P}_{ij}$$

- π_i , from slide 9: the share of its time the walk spends at i .
- $\hat{P}_{ij}$, from slide 5: *given* it is at i , the chance the next event is j .
- Their product: the share of **all** corpus moves that are this one.

Conditional asks *standing here, where next?* **Joint** asks *how much of the music is this?*

Ranking by joint puts **what Parker plays most** on top, not what is most rule-like.



Top 20 transitions ranked by joint probability $\pi_i \cdot \hat{P}_{ij}$. **All 20** stay on the same chord quality. **14** move by a step or less and **13** descend; every one of the other **6** is a leap from one chord tone to another.

Questions I didn't have time to address

- **How long until Parker plays the same thing again?** How often each state comes back round.
- **What does a solo look like backwards?** Whether the language still makes sense played in reverse, and by how much it does not.
- **What does tension look like when it never resolves?** The shape of a line that keeps postponing the landing.
- **How surprising is the next note, on average?** One number for how predictable the whole language is.
- **Does the vocabulary split into neighbourhoods?** Whether the line settles inside a region for a while before moving on.
- **Is resolution reliable or erratic?** The spread of the waiting times, not only their average.
- **How long before the walk has played everything it knows?**

What the operator did and did not recover

What it recovered

- **Voice leading, as the busiest traffic.** Ranked by joint probability $\pi_i \cdot \hat{P}_{ij}$, the twenty most-travelled transitions in the entire operator are all moves **inside a single chord**: fourteen by a step or less, thirteen descending, each of the six leaps chord tone to chord tone. That is what a bebop line is made of, from ranking a matrix by traffic.
- **Gravity, as a waiting time.** A typical state reaches a chord tone on a beat in **4.1 events**; the most unstable states get there in **under 2**. The notes that resolve fastest are the ones theory calls approach tones, and the chain found them without being told which notes clash.
- **And gravity has a destination, not just a delay.** The same system says *which* note it lands on: four of the six fastest-resolving states land **by a semitone on the nearest chord tone**, between **49 %** and **66 %** of the time. Nothing told the model what a semitone is.
- **The on-beat / offbeat asymmetry.** All eight fastest-resolving states sit **offbeat**. Tension is placed off the beat and released on it.

What it did not

- **Markov might not be the best tool.** Approached from the 3rd, that same state behaves differently than approached from the 6th.
- **Much of the structure is metre, not music.** The operator is 97 % empty largely because the beat cannot jump: from &4 the walk goes to b1 69 % of the time. Some of what looks like syntax is bar lines.
- **The sampled lines are not Parker.** Same interval sizes, same rest frequency, no phrases. The engraved staff is the honest version of that: it holds up bar to bar and falls apart across a chorus.
- **Other dimensions are missing.** No register, no duration, no dynamics. The state is a pitch class against a chord, on a beat, and nothing else.

What I would do next

Splitting the operator in two

One operator was estimated over all transitions. Splitting it and comparing the halves is the obvious extension, and a split earns its place only if it conditions on something **not already in the state** and not explained by metre.

- **Within-chord versus across-chord.** Does the line behave differently at a chord change than inside one? The flag exists in the data and was never used.
- **Direction** (9,505 up, 11,530 down) and **step versus leap** (11,226 / 8,441). Both are properties of the transition rather than the source, and both are well balanced. Descending lines already land a chord tone on a strong beat **67 %** of the time against **54 %** ascending, $z = 12.4$, so there is something there.
- **Phrase position:** first note after a rest versus mid-phrase.

Other threads

- **More coordinates:** duration, register, dynamics. None are modelled. Duration and register are already in the data, absolute pitch is stored and only the degree is used. Dynamics are not: these transcriptions carry no dynamic marks at all, so that coordinate needs a performance source rather than the score.
- **Interval vectors.** Walk on the interval content of a short window of the line instead of on one note against one chord. It throws away which note is which and keeps only the distances, so it is a coarser description, but it is the same for a phrase and its transposition, which the current encoding is not.

Thank you

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EXTRA SLIDES

Backup

Material held back from the 20 minutes, kept for questions.

Put all the probability on one state and let it run

Slide 9 found π algebraically, as the left eigenvector at $\lambda = 1$. Here it is found the other way, by iteration:

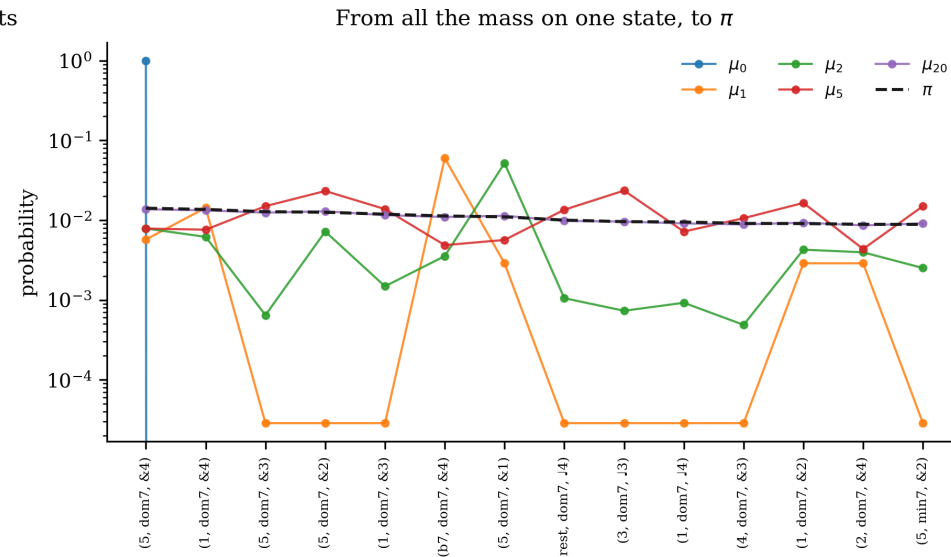
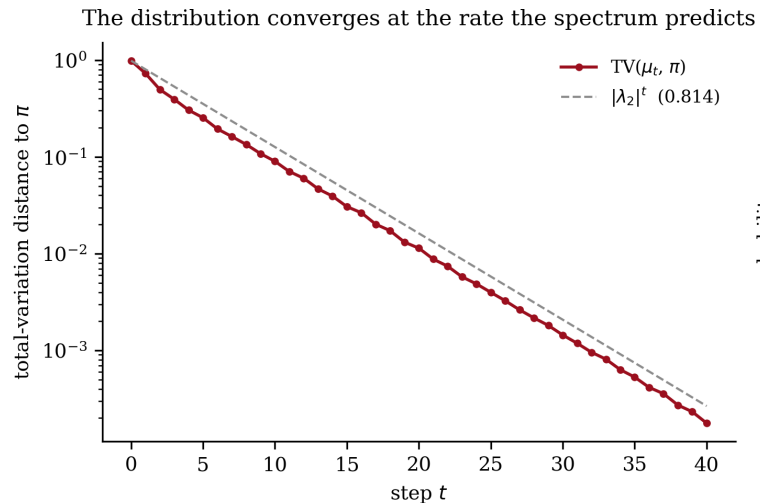
$$\mu_0 = e_i, \quad \mu_{t+1} = \mu_t \hat{P}$$

Start with certainty about the note, and watch the certainty dissolve.

Starting on the busiest state, (5, dom7, &4):

- after **2 steps** the total-variation distance to π is already below 0.5;
- after **21 steps** it is below 0.01;
- at $t = 40$ the largest disagreement with π is 8.7×10^{-6} .

The decay tracks $|\lambda_2|^t$ almost exactly: the mixing time, measured rather than asserted.



The spectrum contains a metric cycle

Every step the walk forgets a little of where it started, and it forgets at a fixed rate: whatever you still know shrinks to **81 %** of itself each step. After five steps about a third is left, after twenty almost nothing.

$$\tau_{\text{mix}} \approx 1 / (1 - |\lambda_2|)$$

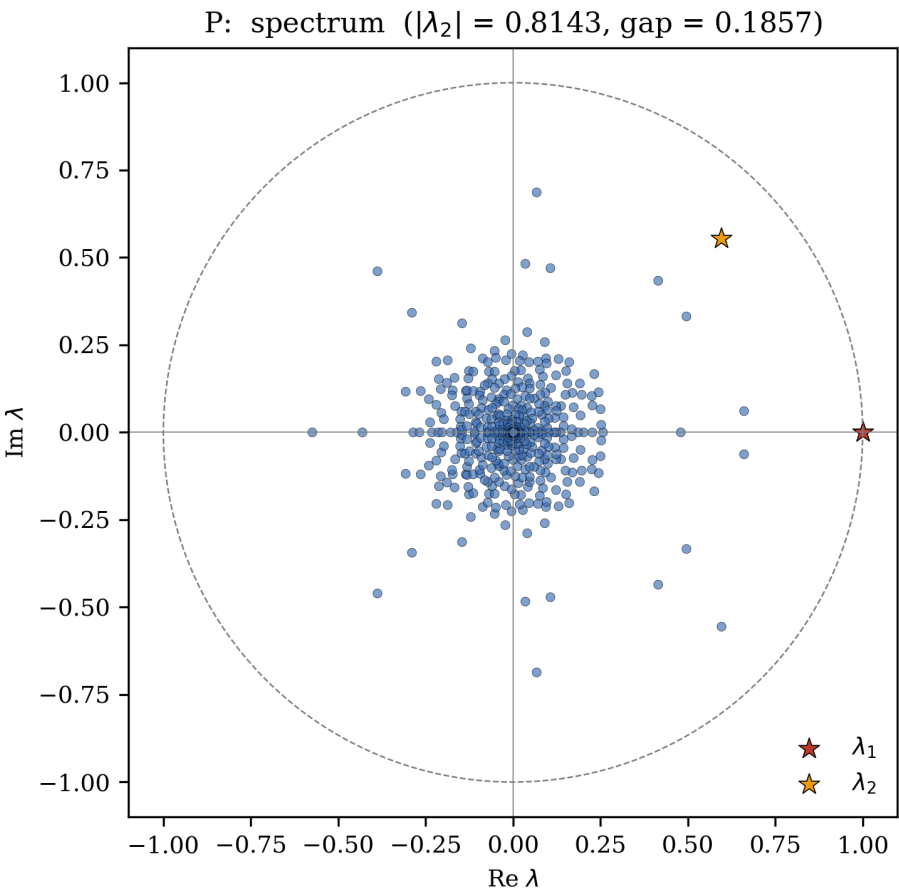
λ_2	0.596 + 0.555 i
$ \lambda_2 $	0.8143
τ_{mix}	5.4 steps

λ_2 is complex, and that is the result

A plain number would mean the walk just fades toward π . A **complex** one means it also *circles*, returning to similar territory at a regular interval:

$$2\pi / \arg(\lambda_2) = 2\pi / 0.750 = 8.4 \text{ events}$$

One bar of eighth notes is exactly 8 events. **The metre is in the operator**, and nothing put it there: it was built by tallying which state follows which.



One dot per eigenvalue. The blob near the centre is short-lived patterns; the star at 1.0 is π , which never decays; the star off the axis is the circling mode, and its angle is the 8.4-event period.

How long until Parker plays this state again

Hitting time looks forward to a set the walk has not reached yet. **Recurrence time** asks the opposite: standing on a state now, how many events pass before the line plays it again.

$$m_i = 1 / \pi_i$$

Kac's formula, so π is the whole answer. No new estimation.

The busiest state, the fifth of a dominant on the pickup, returns every **71 events**, roughly **nine bars** of eighth notes. The median state waits **809**.

So Parker's core vocabulary comes round about once a chorus, and the rest of the state space is genuinely occasional.

Read the right-hand spike with care. Those are states with almost no data, whose π is mostly smoothing. Their recurrence times say more about α than about Parker.

